GRABage: ONLINE WASTE REPORT COLLECTION AND MANAGEMENT SYSTEM

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RATIONALE

The rapid increase in population has led to improper waste management in cities resulting in the increased spreading of diseases and pests. Waste management is necessary today because, with the rapidly growing population, waste generated by humans is also increasing. has its own manual approach to the waste management system, yet it has its limitations. Traditional waste management is using fixed schedules of when and where to collect the waste. Yet, this scheduling was rarely executed because of many reasons, such as human errors and unavoidable change of weather (rain, storm, earthquake, etc.). For those reasons, the researchers came up with making a mobile application and web-based system to notify waste collectors and prioritize barangays with urgent needs.

OBJECTIVES

The general objective of the study is to develop an online waste report collection and management system. The study attempted to identify the problems being encountered in terms of Garbage Collection Schedule and Garbage Collection Monitoring; the features needed to integrate in the proposed system; suitable hardware and software neededto develop the proposed system; proposed system that deliver a consistent result through a set of standardize procedure; level of acceptance of the proposed system based on the evaluation of the IT experts in terms of Functionality, Reliability, Portability, and Security; and level of satisfaction of the respondents of the proposed system in terms of Usability and User Experience.

METHODOLOGY

The researchers used the descriptive developmental research design, and Agile process model to make the process more flexible. In addition, the researchers used the questionnaire for measuring and comparing the level of acceptability and level of satisfaction of the end-users and IT experts on the developed online waste report collection and management system. The researchers used convenience sampling due to the on-going pandemic. The researchers used the Likert Scale to measure the holistic view and opinion to measure the level of acceptance and satisfaction of the study.

RESULTS

Based on the evaluation among respondents, the study yielded the following findings: a. The residents are having difficulties with the existing waste management in getting notifications and monitoring the collection of waste; b. several features are developed to solve the problems encountered by the residents and barangay staff; Truck Tracker and SMS notification, features that helps the waste collector and residents to easily locate the user through real-time location and be easily monitored the distance of the waste collector through SMS notification; c. the IT consultant suggested to developed Validation check in compliance with the data privacy law when asking personal information and consider the future of the system to make it reusable.

CONCLUSIONS

On findings of the study, the following conclusions were drawn; a. The researchers concluded that developed systems can be beneficial to the municipality and its residents who's having difficulties on the proper management scheduling for waste collection; b. there are so much to be done to make the system user-friendly to its customers by delivering precise results of data; c. most of the hardware and software needed are a variety of modern technologies and help the process of the development of the system in the easiest way.; d. the system passed the analysis and can deliver its purpose to the user; e. the system can perform each process based on the given task; f. the system can be utilized as an innovative and effective tool in terms of waste management schedule and garbage collection notification.

KEYWORDS *GRABage, garbage collection, scheduling system, GIS, SMS notification, information system*

1. INTRODUCTION

Every year people dump a massive 2.12 billion tons of waste. If all this waste was put on trucks they would go around the world 24 times. This stunning amount of waste is partly because 99 percent of the stuff we buy is trashed within 6 months [1].

Tons of waste is produced by people all around the world every day. The rapid increase in population has led to improper waste management in cities resulting in the increased spreading of diseases and pests. Waste management is necessary today because, with the rapidly growing population, waste generated by the human is also increasing. Hence, in the absence of an effective waste management system, it may cause health issues and natural disasters in the long run.

In the Philippines, around 35,580 tons of garbage is generated every day. Typically, each person in the country produces about half kilogram of garbage every day in the rural and urban areas, respectively. For Metro Manila, it is estimated that 8,636 tons of solid waste is generated per day, which around 0.7 kg (kilogram) per person per day due to its more modernized lifestyle. The household of our society is the biggest source of garbage in the Philippines at seventy-four (74%). Furthermore, among the total solid waste generated from households, ninety-five (95%) can still be recycled or reused (43%), or turned into compost (52%). Only 5% is made up of residuals (4%) and special/hazardous waste (1%) that are no longer biodegradable or usable [2] . In the country, the LGUs (Local Government Units) hold the primary responsibility for the efficient and effective solid waste management. Despite this law, however, poor solid waste management in the Philippines is still prevailing since controlled and open dumps are being used in the country.

Nowadays, there is existing waste collection management wherein they are using fixed schedules of when and where to collect the waste. As the dump trucks must collect every solid waste in a bin around their designated routes, the truck will be loaded. If the truck is loaded, they are obligated to unload the waste in the dumpsite and may forget to continue the collection of solid waste. So, the problem is the overflowing of neglected wastages on the roads beside the fact that they collect the solid waste rarely in a week or month. The Nasugbu, Batangas has its own manual approach to waste management system, yet it has its limitation. The waste collectors in Nasugbu are using fixed schedules of when and where to collect the waste. Yet, this scheduling was rarely executed because of many reasons, such as human errors and unavoidable change of weather (rain, storm, earthquake, etc).

For that reasons, the researchers proposed an innovation through proposing a GRABage: Online waste report collection and management system. The project is intended to create a mobile application and web-based system to notify waste collectors and prioritize barangays with urgent needs.

2. OBJECTIVES

This study aimed to develop a GRABage: Online waste report collection and management system. Specifically, the study attempts to answers the following question:

1. What are the problems being encountered in terms of:
    - 1.1 Garbage collection schedule
    - 1.2 Garbage collection monitoring
2. What are features needed to integrated in the proposed system?
3. What are suitable hardware and software needed to develop the proposed system?
4. Will the proposed system deliver a consistent result through a set of standardize procedure?
5. What is the level of acceptance of the proposed system based on the evaluation of the IT experts in terms of:
    - 5.1 Functionality;
    - 5.2 Reliability;
    - 5.3 Portability; and
    - 5.4 Security?
6. What is the level of satisfaction of the respondents of the proposed system in terms of:
    - 6.1 Usability; and
    - 6.2 User experience?

3. LITERATURE REVIEW

Based on the study of Idowu et al. (2012) entitled “Development of a Web based GIS Waste Disposal Management System for Nigeria”, developed a web-based solid waste management system for a community of Nigeria that allows residents to view the scheduled time that the garbage will be collected. It shows the system architecture, the person involved and what framework and tools was used in developed system. The system will ease the work or process of waste management of the local government units of Nigeria since they were able to monitor, manage and maintain the location of all the garbage collection tanks within the area. The study

concluded that, the use of system will achieve and ensure the cleanliness of environment and decrease the risk experiencing on improper management of waste of sources in the area of Nigeria [3].

As the population rapidly increasing, the waste produced is also growing at the same time. The country of India implemented various ways on how to solve the problem about garbage collection that is quite hard to solve since the existing system is fortuitously which leads overflowing and smelly garbage due to areas left by the collection of waste. Therefore, a study entitled "Smart Garbage Management System" by Jajoo et al. (2018) proposed a system that monitors the household garbage bin regarding the level filled. An Ultrasonic sensor is used to measure the level of garbage collected in dustbin. When the garbage level reaches a certain level, the system will send a notification to the appropriate authority via various ways. If the bins are already full or nearly full, a warning message will be generated and forwarded to the municipality via notifications, by using the WI-FI module. This will enable them in determining where and when they should go to collect waste. This keeps track of the effort of checking out the area by going there. The system aid both citizens and the Brihan Mumbai Municipal Corporation (BMC) which usually fails to gather the waste of some areas. When a notice is ignored by a lower authority, the following notification is sent to the higher authority. This allows us to efficiently handle waste. Additionally, an Android application is being developed to assist garbage collectors in informing the authorities when waste is being collected. End- to-end communication is established as a result of this [4].

A study entitled "Automated Garbage Collection Alert System" developed a mobile application to manage garbage in various locations based on the analysis of a technique for the optimization of the waste collection and transportation system, which was based on Geographic Information System (GIS). Through this mobile application, it allows communicating directly with the Municipal Council via this smartphone application and receive information about the garbage dump's specific location. It focuses on waste dumps in the streets and main roads in this system. People and local councils in a nation like Sri Lanka can save time and money by utilizing thissmartphone application [5].

4. MATERIALS AND METHODS

The researchers used the descriptive-developmental method in the study. Descriptive research uses a quantitative research method by collecting quantifiable information concerning about the current status of the study. The purpose of using descriptive method is to determine average and frequencies of the data and information gathered. While developmental method is to be used since this study introduced a waste management system named GRABage: Online Waste Report Collection and Management System

The system used the Agile process model in developing and finishing the waste report and management system. The model consists of five cyclical phases: Brainstorm, Design, Development, Quality Assurance and Deployment.

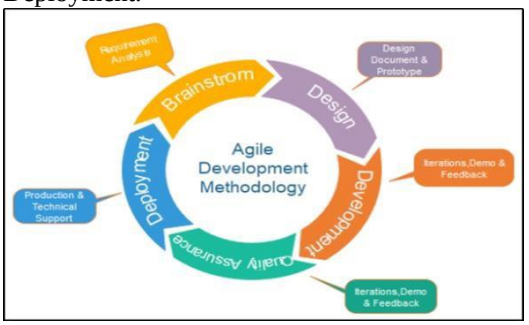

Figure 1. Agile Process Model

Figure 1 shows the Agile Software Development Life Cycle (SDLC). It is an SDLC that focuses on adapting changes that became popular because of its effectiveness and result-oriented approach. This model is suited for the developed system since it focuses on the iterative development.

Brainstorm Phase

The brainstorm phase often referred to as the requirements gathering phase wherein the researchers determined the functions and requirements needed in the study. The researchers also gathered information in order to analyze the business requirements and system deliverables.

Design Phase

In the design phase, visual design and the architectural structure of the app are the two ways to approach this phase. The researchers determined the features of the system and describe operation and flow where it has a goal to be user-friendly.

Development Phase

The development phase is also known as construction stage where writing codes and converting design into actual software occur. The researchers developed the system using programming process, codes and commands, in order to meet business requirements and desire deliverables.

Quality Assurance (QA) Phase

In the quality assurance phase or the testing phase, the researchers will examine the project's performance and making sure that it is bug-free software. Series of test will be conducted and researchers will analyze project to ensure the code is clean and business goals are met.

Deployment Phase

In the deployment phase, the researchers issue the project to the user's work environment – either for demo or actual use. Further iteration happens when there is a new introduces features on the software. The system was evaluated and its effectiveness and quality was assessed.

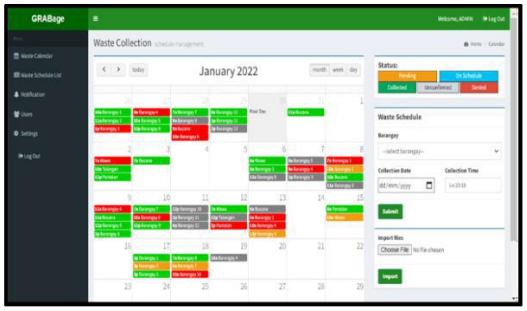

Figure 2. Waste Calendar on Admin Module

Figure 2 displays the waste calendar on the Admin module. On this interface, admin can manage waste schedule. There are five (5) different status of waste collection and it represents different colors. Orange color is for Pending status, blue is for On schedule, the collected status as green, Gray for Unconfirmed status and red as Denied. Schedules can be added in two different ways, importing files to add more than one schedule or setting individual schedule for a specific barangay. On the left side of the screen, navigational menu is added and the menu includes Waste Calendar, Waste Schedule List, Notification, Users, Settings and Logout button.

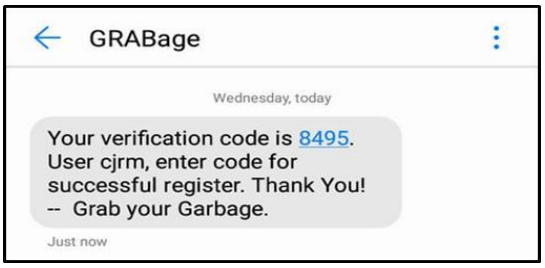

Figure 3. Verification Code on Resident Module

The figure 3 shows the message including the verification code that was sent when resident sign up to GRABage. To successfully register, resident should enter code correctly to the application.

Figure 3. Verification Code on Resident Module

As shown in the figure 4, residents will be able to capture image to send request for prioritization of their barangay and the image will serve as proof for faster verification of the request.

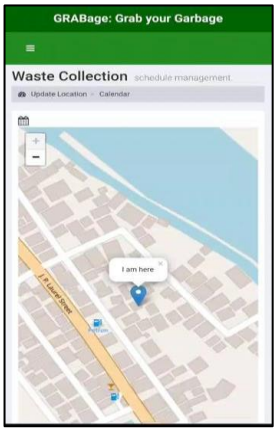

Figure 5. GPS Location on Resident and Staff Module

Figure 5 displays the real-time location of the residents and staff when "Update Location" button is clicked. It shows the specific location of the user using map. Beside that button, "Calendar" button is located to view detailed waste schedule of their barangay.

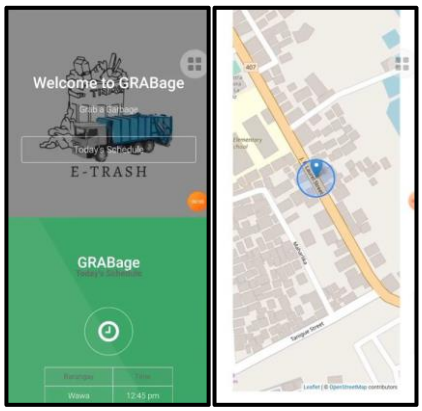

Figure 6. Truck Tracker

On the figure 6, it shows the module for truck tracker. The left image shows the truck waste collector that can view barangays scheduled for today's collection. When the truck started collecting waste, real-time location of the truck is display on the application, as shown on the right image.

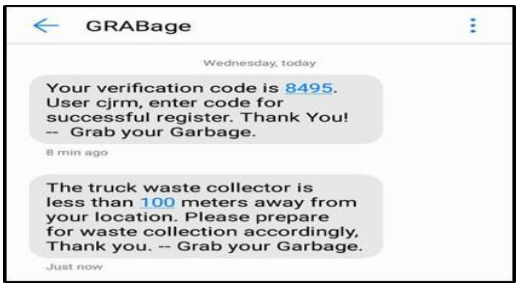

Figure 7. SMS Notification

For the figure 7, it displays the message that was sent to users with today's schedule and the truck waste collector is less than 100 meters away from their location. The purpose of this message was to notify users about the distance of the truck and they can have time to collect and gather their waste and to pile outside while waiting to be collected.

Hardware and Software Requirements

Reflected on the Table 1 and 2 are the hardware and software needed in developing the GRABage: Online Waste Report Collection and Management System.

Table 1. Hardware Requirements

Particulars	Specification
Processor	2.30 GHz
RAM	4 GB RAM
Storage	500 GB
HDD	500 GB or more
Input Devices	Keyboard/Mouse
Output Devices	Monitor

Table 2. Software Requirements

Particulars	Software
Operating System	64-bit Operating System
Database	MySQL
Programming Language	PHP
Web Editor	XAMPP
Web Technology	HTML, CSS

5. RESULTS AND DISCUSSION

In the preparation and evaluation stage, the researchers conducted the evaluation procedure of the developed system to the fifty (50) respondents comprising of thirty (30) residents of Nasugbu, fifteen (15) Barangay Staff and five (5) IT Experts. The survey questionnaires made by the researchers were validated and used to answer the following question.

The researchers provided two (2) sets of survey questionnaires, one for the end users (barangay staff and residents) that seek to answer the evaluation forms to determine problems being encountered in terms of garbage collection schedule and garbage collection monitoring and also to determine level of satisfaction in terms of Usability and User Experience which is conducted personally and another set of questionnaires through Google forms for the IT experts that seek to answer the evaluation forms to determine level of acceptance of the developed system in terms of Functionality, Reliability, Portability and Security. Also, they were asked to answer the evaluation forms to determine level of satisfaction of the end users of the developed system in terms of Usability and User Experience.

Table 1. Problems Encountered by the End users Relative to the Existing Garbage Collection Schedule

Problems	Frequency	Rank
Schedules are not being followed because of waste truck collectors are limited.	36	2
There is no rescheduling method after the waste collectors failed to collect waste from certain location.	41	1
Users are not updated when there are changes in the waste collection schedule.	30	3

Table 1 presents the problem encountered by the end users relative to the existing garbage schedule. Based on the result stated on the table above, "There is no rescheduling method after the waste collectors failed to collect waste from certain location" got the highest rank. The researchers find that the users aren't able to request another schedule

when the waste collectors failed to collect. The user must be able to request rescheduling to avoid their waste piling up and being neglected. Related to the study of Idowu et al. (2012) entitled "Development of a Web based GIS Waste Disposal Management System for Nigeria" that focuses in providing residents the scheduled time that the garbage will be collected to decrease the risk of improper management of waste.

Table 2. Problems Encountered by the End users Relative to the Existing Garbage Collection Monitoring

Problems	Frequency	Rank
Collected waste is not monitored by MENRO.	36	2
Unmonitored overflowing of garbage in barangay results unpleasant pile of waste in roads and bins.	37	2
Detection and management of waste is one of the primary problems of the municipality.	42	1

Table 2 shows the problems encountered by the end users relative to the existing garbage monitoring. Based on the table above, "Detection and management of waste is one of the primary problems of the municipality" got the highest rank. This clearly states that one of the primary problems of the municipality is the detection and management of waste. Most of barangay relay on municipal waste collector. So, if the waste of different barangays isn't collected it might cause some issues such as health issues or piling of waste on roads and bins. That is also the reason why the India created the "Smart Garbage Management System", studied by Jajoo et al. (2018), that propose a system that monitors the household garbage bin regarding the level of waste filled.

Table 3. Features Needed to Integrate in the Developed System

Feature	Function	Description
Calendar	The Calendar is added to help the admin to easily see the schedule for the month, week, or day. With this, the admin can be easily aware about the status collection at a certain time.	A series of pages or charts of days, weeks, months, or specific years. This may contain schedules and holidays.
Verification Code	Through the help of SMS notification, this feature sends randomly generated numbers to the user's personal number. This helps the application to be more secure and reliable when signing up to the system.	The Verification code is an OTP (One Time Pin) that is randomly generated to let users have security protection.
Image Capturing	The Camera Capture feature is designed to capture the report of end-users, reports like piling trash or uncollected waste in their barangay. This report will be used to let the residents request for collection on their place. This will allow the admin to identify the validity of the requests of the users.	Image capturing is an act of recording and/or capturing an image or other information using a computer or camera. This is a technology designed for capturing visual images.
GPS location	The GPS location feature is added on residents' modules. This feature updates the location of the user through a button, and then will also be used for waste collection routes.	A GPS Location is widely used to navigate someone's exact location anywhere around Earth's surface.
Truck Tracker	The truck tracker is a real-time location feature that helps the end users to see the location of the truck and with the integration of SMS notifications, residents get notified when truck waste collector is 100 meters away from their location.	Truck Tracker is used to keep track of the real-time location of a specific truck.
SMS Notification	Not all users have the most advanced smartphones. The SMS Notification feature of our developed system made it possible to let end users is updated about the truck collector's distance away from on-scheduled barangay.	A SMS Notification is a short message sent in response to an event, can be an update or service alert.

Table 3 shows the features needed to integrate to the developed system and their corresponding descriptions on how it is used in the system. Moreover, these features help the researchers to achieve the functionality and purpose that the system should deliver to its customers. The most highlighted features in the system are the truck tracker and SMS notification and it is also highlighted on the study of Ghazali et al. (2019) about "Mobile Application on Garbage Collector Tracker Using Google Maps" that the project was designed to alert communityon garbage duty schedule, be notified and be able to track the location of garbage collector using Google maps application [6].

The following are the hardware needed to run the web application.

Table 4. Hardware Specification for Web Application

Items	Requirement	Description
Processor	2.30 GHz	A better processor can process basic command and can help in better performance of the system.
Memory/ RAM	4.00 GB	For the better speed and performance of our application, the researchers require at least 4GB RAM
Storage	500 GB	The researchers require at least 500GB of storage capacity to ensure the best performance of the application
Input device	Keyboard, Mouse	The input devices help the user to input their basic and required information. The input devices can help in better processing of data.
Output device	Monitor	The output device/s can help in producing better way to let end users interact through the application
Communication Interface	Mobile device, Modem	The communication interface helps the end users to have better communication.

Table 4 shows the hardware needed to run the web application. It also shows the requirement specification and its description of use of each hardware. To acquire better performance of the web application, a better processor, higher storage and at least 4 GB memory are recommended by the researchers. To avoid hassle using the system, researchers suggested to use keyboard and mouse as input devices and a monitor as an output device. And also for easiest access to the web application, mobile device and modem are suggested as communication interface.

The following are the hardware needed to run the mobile application.

Table 5. Hardware Specification for Mobile Application

Items	Requirement	Description
Processor	1.0 GHz	A better processor can process basic command and can help in better performance of the system.
Memory/ RAM	2.0 GB	For the better speed and performance of our application, the researchers require at least 4GB RAM
Storage	16 GB	The researchers require at least 500GB of storage capacity to ensure the best performance of the application
Display	5.5 inches, 720 x 1280 pixels	A better display can help to produce better way to let end users interact through the application
Communication Interface	Mobile device, Modem	The communication interface helps the end users to have better communication.

Table 5 shows the hardware needed to run the mobile application. It also shows the requirement specification and its description of use. To acquire better performance of the mobile application, a better processor, higher storage and at least 2 GB memory are recommended by the researchers. To avoid hassle using the system, researchers suggested to use a 5.5 inches, 720 x 1280 pixels for better display and interaction between end user and mobile application. And also for easiest access to the mobile application, mobile data and modem as communication interface.

The following are the software needed to run the web application.

Table 6. Software Specification for Web Application

Items	Requirement	Description
Web Browser	Google Chrome, Microsoft Edge	Using the required web browser can have a better percentage of the performance success
Operating System	64bit OS, Android version	The required operating system can help the end users to enjoy the technology feature of the application such as Real-Time Location and SMS Notification.

Table 6 shows the recommended software for the implementation of the developed system. To acquire smooth interaction of web application and end user using their personal devices, 64bit OS and android version is recommended to enjoy the major features of the application. And for better performance, the required web browser such ass Google Chrome and Microsoft Edge is recommended.

Table 7. Software Specification for Mobile Application

Items	Requirement	Description
Operating System	Android Lollipop above	The required operating system or above can help the end users to enjoy the technology feature of the application such as Real-Time Location and SMS Notification.

Table 7 shows the recommended software for the implementation needed to run the mobile application. The mobile should be Android Lollipop above to smoothly run the system on mobile devices and can definitely enjoy the major features of the application.

Table 8. Results from Quality Assurance Analysis

Module	Scenario	Comments	Result
Admin Module	Login	Security services must be develop to prevent unauthorized access to, modification of private date (data).	Passed
	Adding of inputs like dates, time and reason for declines	The system should be able to handle negative inputs and testing of data.	Passed
	Notification	Notifications should notify the admin in all status even offline	Failed
	Prioritization Request	Set cut-off time to provide reminders on the inquiry logged after the cut-off time that this will be entertained on the next working day to avoid delays and backlogs	Failed
Staff and Resident Module	End users inputs private information	Provide protection for all gathered private information of all end users.	Passed
	Reminder and Tips	Provide guidelines, rules and regulation that both parties will use the software properly. Give assurance between waste management and residents about the maintenance, updates and future of the system.	Failed

Table 8 shows the components tested by the Quality Assurance (QA) analyst which are the admin module, staff and resident module and the truck tracker application. Using a Test Case Document and video presentation, QA analyst discovers bugs and errors that might not satisfy users' requirements. Scenarios like login, adding inputs like dates passed the evaluation on the admin module and entering private information for the staff and residents. However, failed results encountered in notification and prioritization request on the Admin Module. Notification should be able to notify admin even offline and setting of cut-off time to be entertained from the residents. On the Staff and Resident Module, failed results include the reminder and tips. This should include guidelines and regulations on how to use the system.

Table 9. Level of Acceptance of the Developed System based on the Evaluation of the IT Experts

Parameters	Weighted Mean	Verbal Interpretation
Functionality	4.60	Perfectly Acceptable
Reliability	4.65	Perfectly Acceptable
Portability	4.80	Perfectly Acceptable
Security	4.70	Perfectly Acceptable
Combined Weighted Mean:	**4.71**	**Perfectly Acceptable**

Table 9 shows level of acceptance of the developed system based on the evaluation of IT Experts. Based on the result of the evaluation, portability got the highest weighted mean and gives conclusion to the researchers about how does the system is portable and can adapt to different kinds of environment. It is supported with the study of "Web-Based Toxic Waste Management Information System Using Unified Approach Method" that accelerates the method of manual recording and reporting of waste management and includes the benefits of the system in overcoming the existing management of waste [7].

Table 10 Level of satisfaction of the end users to the developed system

Parameters	Weighted Mean	Verbal Interpretation
Usability	4.60	Extremely Satisfied
User Experience	4.79	Extremely Satisfied
Combined Weighted Mean:	**4.70**	**Extremely Satisfied**

Table 10 shows the level of satisfaction of the end users to the developed system in terms of Usability and User Experience. Based on the result, end users are extremely satisfied to the developed system. This means that the users easily get familiar with the developed system and to learn how to use it properly. It is

supported with the study of Idowu et al. (2012) entitled "Development of a Web based GIS Waste Disposal Management System for Nigeria", that aims to ease the experience of the end users while performing great usability.

Table 11. Level of satisfaction of the IT Experts to the developed system

Parameters	Weighted Mean	Verbal Interpretation
Usability	5.00	Extremely Satisfied
User Experience	4.76	Extremely Satisfied
Combined Weighted Mean:	**4.88**	**Extremely Satisfied**

Table 11 shows the level of satisfaction of the IT Experts to the developed system in terms of Usability and User Experience. Based on the result, IT Experts are extremely satisfied to the developed system.

6. CONCLUSIONS

This part shows all the conclusions that the researchers gathered during the development and evaluation of GRABage.

1. Based on the findings, the researchers concluded that GRABage application can be beneficial to the municipality and its residents having difficulties on the proper management scheduling for waste collection.
2. Given the different features implemented to the developed system, researchers have concluded that (1) Image Capturing can be beneficial to prove the request of the residents about the prioritization, (2) Calendar that allows admin to create manageable schedule of waste collection, (3) Verification code that ensure securities for those who are signing up in the system, (4) GPS Location that the residents can use to save their location (5) Truck tracker which is to track the distance of the truck to the residents and it is notified through (6) SMS notification.
3. The researchers have concluded that most of the hardware and software needed to integrate to the developed system are a variety of modern technologies and helps the process of the development of the system in easiest way.
4. The researchers concluded that in terms of main features implemented, the system passed the analysis. For this, it is concluded that the system can deliver its purpose to the user which is to manage schedule for waste collection. However, the system fails on providing the assurance to the future of the system in terms of handling notifications, prioritization request and the reminder and tips interface. With this, improvements should be added like how notification should be able to notify admin even offline and setting of cut-off time to be entertained from the residents' sent prioritization request and for the reminder and tips, this should include guidelines and regulations on how to use the system.
5. Given the high ratings and evaluation of acceptance of the IT Experts to the developed system, the researchers have concluded that the quality of the developed system conforms to the ISO/IEC 25010 standards. Researchers also concluded that the system can perform each processes based on the given task.
6. With the high ratings of both the end users and the IT experts on the level of satisfaction to the developed system, the researchers concluded that the system can be utilized by the local unit of Nasugbu as an innovative and effective tool in terms of waste management schedule and garbage collection notification.

5. RECOMMENDATIONS

The following are the recommendations for the enhancement of the GRABage: Online Waste Report Collection and Management System.

1. The developed system is recommended by the researchers to be useful to the municipality to improve waste management collection in different barangays preventing one of the main problem which is managing and disposal of waste to ensure that garbage scheduling and monitoring are achieve in faster and easier way.
2. The integration of Heatmap to view barangays that have the most waste per collection and improvements for dashboard to have weekly or monthly generation of data displayed are highly suggested to be included on the developed system.
3. The developed system should provide step by step procedure or tutorial of how to use the system especially for new users.
4. The inclusion of proper presentation of data in printing of reports is also recommended to identify the formality of the documents.

5. For better handling of reports in tables, the developed system should be able to sortstatus on their choice.
6. The GRABage application should have an IOS and Android application that can be easily downloaded from the App Store and Play Store.
7. The research team recommended on giving assurance that the developed system should be reusable in the future and maintain connection between the waste managementand resident users.
8. For future researchers, the research team recommend to enhance the system, make it appealing to the users and more user-friendly.

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